SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – INDUSTRY PROBLEM SOLVING TASK

Course Code:	CSA12	**Course Title:**	Computer Architecture
CO Assessed:	CO4 – Precision (BL4)	**Assessment:**	Industry Problem solving task
Weightage:	30%	**Total Marks:**	100
CO4 Statement:	Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.		
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Section / Batch:	CSE(AI) / 2ND YEAR	**Date:**	17/08/2026

Instructions to Students

- Identify the relevant concepts of cache hierarchy, SRAM, DRAM, MRAM, NAND Flash, RRAM, memory latency, bandwidth, and virtual memory paging.
- Analyze the memory-access requirements of smartphones, cloud servers, autonomous vehicles, edge AI devices, and gaming consoles, considering latency, bandwidth, capacity, and power consumption.
- Apply suitable memory-hierarchy techniques such as cache optimization, appropriate memory technology selection, data locality, paging optimization, and hierarchical memory organization.
- Compute performance measures such as memory access time, latency, bandwidth, throughput, speedup, hit/miss rates, and resource or power utilization where applicable.
- Interpret the results by evaluating performance improvements, memory bottlenecks, technology trade-offs, and the suitability of the proposed memory hierarchy for each application.

QUESTIONS

1. A large-scale e-commerce platform experiences slow response times during peak shopping periods when thousands of users simultaneously access product catalogs, customer records, and transaction data. Analyze the roles of cache memory, DRAM, and secondary storage in handling these workloads. Propose a memory hierarchy that reduces access latency, improves data locality, and supports high concurrent transaction throughput.
2. A robotic manufacturing system must process camera images, sensor readings, and control instructions with strict real-time deadlines. Evaluate the suitability of SRAM, DRAM, and non-volatile memory for storing frequently accessed control data, sensor information, and program code. Design a memory hierarchy that minimizes latency and ensures predictable memory access for real-time operations.
3. A video-editing workstation experiences delays while processing multiple streams of high-resolution 4K and 8K video. Analyze how cache size, DRAM capacity, memory bandwidth, and storage performance influence video-processing workloads. Recommend a hierarchical memory organization that improves data transfer rates, reduces processing delays, and supports smooth real-time editing.
4. A weather forecasting system performs complex numerical calculations on massive datasets containing atmospheric temperature, pressure, wind, and humidity measurements. Analyze how cache hierarchy, DRAM bandwidth, and large-capacity memory influence the performance of repeated numerical computations. Propose memory-hierarchy enhancements that improve data locality, reduce memory-access bottlenecks, and increase computational throughput.
5. A smart home gateway manages data from cameras, motion sensors, environmental sensors, and connected appliances while operating continuously with limited energy resources. Compare SRAM, DRAM, Flash, and other suitable memory technologies based on latency, capacity, persistence, and power consumption. Design an efficient memory hierarchy that enables rapid access to frequently used data while minimizing energy consumption and maintaining reliable long-term data storage.

Code of Conduct

I LOKESH S (192572151) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: LOKESH S

RUBRICS FOR CALCULATION

Criteria	Weightage(%)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginnin (0–1 Mark)
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Marks Evaluation:

Q. No	Understanding of Industry Problem (20)	Application of Engineering Concepts (25)	Solution Design (25)	Use of Tools (15)	Analysis (10)	Professional (5)	T (out
1	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 25	/ 25	/ 15	/ 10	/5	

Course Coordinator **Faculty Incharge**

1. E-Commerce Platform – Memory Hierarch

A large e-commerce platform has to serve thousands of users simultaneously. During peak shopping periods, users continuously access product details, customer information, inventory, and payment transactions. A proper memory hierarchy is required to reduce response time.

Role of Memory Components
Cache Memory

- Cache is the fastest memory near the CPU.
- Frequently accessed product details, database indexes, session data, and recent transactions can be stored in cache.
- It reduces repeated access to slower main memory.
- L1 cache is the fastest and smallest, followed by L2 and L3.
- A large shared L3 cache is useful when many processor cores handle concurrent users.

DRAM

- DRAM acts as the main memory.
- It stores active databases, application data, user sessions, query results, and transaction buffers.
- Large DRAM capacity reduces the need to read data repeatedly from storage.
- High memory bandwidth allows several CPU cores to access data simultaneously.

Secondary Storage

- SSD or NVMe storage provides permanent storage.
- Product databases, customer records, transaction history, images, and logs are stored here.
- NVMe SSDs provide much faster access than traditional HDDs.
- HDDs may still be used for backups or historical data.

Proposed Memory Hierarchy
CPU Registers → L1 Cache → L2 Cache → Shared L3 Cache → Large DRAM → NVMe SSD → HDD/Cloud Backup

Performance Improvements

- Store frequently accessed product information in cache.
- Keep active database tables and indexes in DRAM.
- Use database caching for commonly executed queries.
- Use NVMe SSDs for fast database reads/writes.
- Apply prefetching to load related product information before it is requested.
- Use multi-channel DRAM to improve bandwidth.
- Maintain temporal locality by keeping recently accessed data in cache.
- Maintain spatial locality by storing related product records close together.

Conclusion
A multi-level cache combined with large, high-bandwidth DRAM and fast NVMe storage reduces access latency and supports thousands of concurrent transactions efficiently.

2. Robotic Manufacturing System – Real-Time Memory

A robotic manufacturing system processes camera images, sensor readings, and control instructions. Since robots must respond immediately, predictable and low-latency memory access is important.

SRAM
SRAM is extremely fast and does not require refreshing.
Suitable for:

- Motor control parameters
- Frequently accessed sensor values
- Interrupt data
- Real-time control variables
- Critical buffers

Advantages:

- Very low latency
- Predictable access time
- Suitable for real-time operations

Disadvantages:

- Expensive
- Low capacity
- Higher area requirement

DRAM
DRAM provides much larger capacity at lower cost.
Suitable for:

- Camera frames
- Image-processing data
- Large sensor buffers
- Temporary computation data

Advantages:

- Large capacity
- Lower cost per bit

Disadvantages:

- Slower than SRAM
- Requires refresh operations
- Access latency is less predictable.

Non-Volatile Memory
Flash, EEPROM, or similar non-volatile memory retains information even when power is removed.
Suitable for:

- Robot firmware
- Program code
- Configuration information

Its access speed is slower than SRAM and DRAM, so critical real-time data should not depend directly on Flash during operation.

Proposed Memory Hierarchy

CPU Registers → L1/L2 SRAM Cache → On-Chip SRAM → DRAM → Flash/NVM

Data Placement

Data	Suitable Memory
Control instructions	SRAM/Cache
Motor parameters	SRAM
Current sensor readings	SRAM
Camera frames	DRAM
Image-processing buffers	DRAM
Program code	Flash/NVM
Configuration	Flash/EEPROM

Real-Time Improvements

- Lock critical instructions and data in SRAM/cache.
- Avoid cache misses for important control loops.
- Use DMA to transfer camera data directly to DRAM.
- Use separate memory areas for real-time and non-real-time tasks.
- Use deterministic memory scheduling.
- Store frequently used control code in on-chip memory.

Conclusion

SRAM should handle critical control operations, DRAM should handle large camera and sensor datasets, and non-volatile memory should store firmware and permanent configuration. This provides fast and predictable real-time execution.

3. Video-Editing Workstation – 4K/8K Processing

Video editing requires large amounts of data to be transferred and processed continuously. Multiple 4K and 8K video streams can create heavy pressure on cache, RAM, memory bandwidth, and storage.

Cache Size

Cache stores recently accessed video-processing instructions and data.
A larger L2 and L3 cache can:

- Reduce repeated DRAM access.
- Improve video decoding and effects processing.
- Store frequently reused image blocks.
- Improve CPU performance.

However, entire 4K/8K video frames usually cannot fit in cache.

DRAM Capacity
Video editing applications require large amounts of RAM.
Large DRAM capacity allows:

- Multiple video frames to remain in memory.
- Multiple applications to run simultaneously.
- Faster rendering.
- Reduced use of virtual memory.

For professional 4K/8K editing, large-capacity RAM is highly beneficial.

Memory Bandwidth
High-resolution video transfers huge amounts of data.
High bandwidth:

- Transfers frames faster.
- Supports CPU and GPU simultaneously.
- Reduces delays during effects and rendering.
- Allows multiple video streams to be processed.

Multi-channel high-speed DDR memory improves performance.

Storage Performance
Traditional HDDs may become bottlenecks when editing large video files.
NVMe SSDs provide:

- High sequential read/write speed.
- Faster video loading.
- Faster timeline scrubbing.
- Faster project and cache access.

Recommended Hierarchy
CPU/GPU Registers → L1 Cache → L2 Cache → L3 Cache → High-Capacity High-Bandwidth DRAM → NVMe SSD → HDD/External Storage

Recommended Organization

- Cache: instructions and reusable frame blocks.
- DRAM: active video frames and project data.
- NVMe SSD: current video files, scratch disk and temporary rendering files.
- HDD/large SSD: archived video projects and backups.

Additional Improvements

- Use GPU VRAM for GPU-based video effects.
- Use DMA for fast device-memory transfers.
- Use multiple NVMe drives for source files and temporary cache.
- Prefetch upcoming frames.
- Store consecutive video frames sequentially to improve spatial locality.

Conclusion
Large cache, sufficient DRAM capacity, high memory bandwidth, fast GPU memory, and NVMe storage significantly reduce processing delays and enable smooth real-time 4K and 8K video editing.

4. Weather Forecasting System – Numerical Computation
Weather forecasting systems process huge datasets containing temperature, pressure, wind speed, humidity, and other atmospheric measurements. These computations repeatedly access large arrays and matrices.

Cache Hierarchy
Modern processors contain:
L1 Cache
- Very small and fastest.
- Stores immediately required instructions and data.

L2 Cache
- Larger than L1.
- Holds frequently accessed data blocks.

L3 Cache
- Much larger and often shared among multiple CPU cores.
- Useful for large scientific datasets.

A good cache hierarchy reduces repeated DRAM access.
Importance of DRAM Bandwidth
Weather forecasting algorithms frequently perform operations on large arrays.

If memory bandwidth is low:
- CPU cores wait for data.
- Computational units remain idle.
- Overall execution becomes slower.

High-bandwidth multi-channel DRAM can feed data to multiple CPU cores simultaneously.
Importance of Large Memory Capacity
Weather datasets may contain billions of data points.

Large DRAM capacity allows:
- More atmospheric data to stay in main memory.
- Reduced secondary-storage access.
- Larger simulations.
- Faster repeated calculations.

Proposed Memory Hierarchy
CPU Registers → L1 Cache → L2 Cache → Shared L3 Cache → High-Bandwidth DRAM → NVMe SSD → Large Storage System

Memory-Hierarchy Enhancements
1. Cache Blocking
Large matrices are divided into smaller blocks that fit inside cache.
This increases data reuse and improves locality.
2. Prefetching
Future data is loaded into cache before the CPU requests it.
3. Sequential Data Layout
Related temperature, pressure, wind and humidity values are stored close together.
This improves spatial locality.
4. High-Bandwidth Memory

HBM or high-speed DDR memory can supply data to processors much faster.
5. Parallel Memory Channels
Multiple memory channels increase simultaneous memory transfers.
6. NUMA-Aware Allocation
On multi-processor systems, data should be stored near the processor that uses it.
7. Fast NVMe Storage
Used for loading large weather datasets and saving simulation results.

Conclusion

A large multi-level cache, high-bandwidth DRAM, large memory capacity, intelligent prefetching and cache-blocking techniques reduce memory bottlenecks and significantly improve weather simulation throughput.

5. Smart Home Gateway – Energy-Efficient Memory
A smart home gateway continuously receives data from cameras, motion detectors, temperature sensors, smart appliances and other IoT devices. The system must provide fast processing while consuming low power

Comparison of Memory Technologies

Memory	Speed	Capacity	Persistent	Power	Suitable Use
SRAM	Very high	Low	No	Moderate	Cache/control data
DRAM	High	High	No	Moderate/High	Active sensor/video data
Flash	Low/Medium	Very high	Yes	Low	Firmware/logs
EEPROM	Slow	Low	Yes	Low	Settings/configuration
SSD/eMMC	Medium/High	Very high	Yes	Moderate	Camera recordings

SRAM
SRAM provides extremely fast access.
Used for:
- CPU cache
- Frequently accessed sensor information
- Network buffers
- Control variables

Because SRAM is expensive, only small amounts should be used.

DRAM
DRAM can store:
- Camera frames
- Temporary sensor data
- Running applications
- Communication buffers

DRAM provides large capacity but consumes additional power because it must be refreshed.

Flash Memory
Flash is non-volatile.
Used for:
- Operating system
- Firmware
- Application code
- Long-term sensor logs

It consumes little standby power.

EEPROM
Suitable for small permanent information such as:

- Wi-Fi settings
- Device IDs
- Calibration data
- User configurations

Proposed Memory Hierarchy
CPU Registers → SRAM Cache → Low-Power DRAM → Flash/eMMC/SSD → Cloud/External Storage

Efficient Operation
Frequently accessed data such as current sensor values should remain in SRAM/cache.
Active camera frames and temporary IoT data should be stored in DRAM.
Firmware, configuration and long-term logs should be stored in Flash.
Older camera recordings may be moved to SSD, SD card or cloud storage.

Energy-Saving Techniques

- Put unused DRAM banks into low-power mode.
- Use cache to reduce DRAM accesses.
- Use low-power DDR memory such as LPDDR.
- Batch sensor writes instead of continuously writing to Flash.
- Turn off unused memory components.
- Store only important camera events instead of continuous video where appropriate.

Conclusion
An energy-efficient smart home gateway can use SRAM for frequently accessed data, low-power DRAM for active processing, Flash for firmware and persistent information, and SSD/eMMC or cloud storage for long-term data. This hierarchy provides fast operation, low power consumption and reliable storage.